

Rafid Mahmood

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Ottawa, Ontario, Canada

Email: mahmood@telfer.uottawa.ca
Homepage: <http://rafidrm.github.io>

Employment

University of Ottawa, Telfer School of Management

Assistant Professor 2023–pres.

NVIDIA Corporation

Senior Research Scientist 2022–pres.

AI Resident Researcher 2020–2022

Education

University of Toronto, Mechanical and Industrial Engineering

Ph.D Industrial Engineering 2015–2020

Vector Institute for Artificial Intelligence Postgraduate Affiliate 2019–2020

University of Toronto, Electrical and Computer Engineering

M.A.Sc. Electrical Engineering 2013–2015

Honors B.A.Sc. Electrical Engineering 2008–2013

Publications ¹

Pre-prints and Working Papers

- Noise-Aware Demonstration Selection for Fairness-Sensitive Medical In-Context Prediction**
M. Vahabi^{*†}, S. Rahimiaghdam, R. Mahmood, and C. L. F. Sun
under review, 2026.
- Test-Time Coverage: Test-Conditioned Curation for Deployment-Aware Learning**
N. Chang^{*}, M. Shen^{*}, S. Dao, J. Wang, J. Chen, T. Breuel, P. Molchanov, R. Mahmood, and J. M. Alvarez
under review, 2026.
- Training and Deploying Machine Learning Services under the Neural Scaling Law**
Y. Zheng^{*}, R. Mahmood, and N. Chen
working paper, 2026.
- Minority Reports: Balancing Cost and Quality in Ground Truth Data Annotation**
H.-W. Liao^{*†}, C. Klugmann, D. Kondermann, and R. Mahmood
working paper, 2026.

¹Articles that were published with alphabetical author ordering are denoted with $\alpha-\beta$. The primary author is denoted with ^{*}. Supervised students are denoted with [†].

Published Journal Articles and Conference Proceedings

5. **Deep Learning-Assisted Appointment Scheduling Under Uncertainty**
A. Moosavi*, O. Ozturk, R. Mahmood, and J. Patrick
accepted at INFORMS Journal on Data Science (IJDS), 2026.
6. **Position: Stop Reactively Patching Your Model Every Time and Start Proactive Test-Driven AI Development**
N. Chang, M. Shen, J. Wang, R. Mahmood, and J. M. Alvarez
International Conference on Machine Learning (ICML), 2026.
7. **Uncertainty Estimation for 3D Object Detection via Evidential Learning**
N. Durasov*[†], R. Mahmood, J. Choi, M. T. Law, J. Lucas, P. Fua, and J. M. Alvarez
CVPR Workshop on Emerging Directions in Data for Multimodal Foundation Models, 2026.
8. **Scaling-Aware Data Selection for End-to-End Autonomous Vehicles**
T. Dimilioglu*[†], N. Chang, M. Shen, R. Mahmood, and J. M. Alvarez
Computer Vision and Pattern Recognition (CVPR), 2026.
– Highlight ($\approx$ top 10% of accepted papers).
9. **Routing, Cascades, and User Choice for Large Language Models**
R. Mahmood*
International Conference on Learning Representations (ICLR), 2026.
10. **AutoScale: Automatic Prediction of Compute-optimal Data Compositions for Training LLMs**
F. Kang*, Y. Sun, B. Wen, S. Chen, D. Song, R. Mahmood, and R. Jia
Conference on Language Modeling (COLM), 2025.
11. **Optimizing Data Collection for Machine Learning**
R. Mahmood*, J. Lucas, J. M. Alvarez, S. Fidler, and M. T. Law
Journal of Machine Learning Research (JMLR), 26 (38), 1–52, 2025.
– Preliminary version at NeurIPS 2022.
12. **Inverse Optimization: Theory and Applications**
 α - β T. C. Y. Chan, R. Mahmood*, and I. Y. Zhu*
Operations Research (OR), 73 (2), 1046–1074, 2025.
13. **Learning to Optimize Contextually Constrained Problems for Real-Time Decision Generation**
 α - β A. Babier, T. C. Y. Chan, A. Diamant, and R. Mahmood*
Management Science (MS), 71 (2), 1165–1186, 2025.
14. **Can Large Vision-Language Models Correct Semantic Grounding Errors By Themselves?**
Y.-H. Liao*, R. Mahmood, S. Fidler, and D. Acuna
Computer Vision and Pattern Recognition (CVPR), 14667–14678, 2025.
15. **Got (Optimal) Milk? Pooling Donations in Human Milk Banks with Machine Learning and Optimization**
 α - β T. C. Y. Chan, R. Mahmood, D. L. O'Connor, D. Stone, S. Unger, R. K. Wong*[†], and I. Y. Zhu
Manufacturing & Services Operations Management (M&SOM), 27 (6), 1721–1739, 2025.
– First Place for Pierskalla Best Paper Award.
– Finalist for MSOM 2023 Practice-Based Research Competition.
– Runners' Up (second place) for POMS 2023 College of Healthcare Operations Management (CHOM) Best Paper Award.

- **Runners' Up (second place) for INFORMS 2024 Innovative Applications in Analytics Award (IAAA).**
 - **Finalist for INFORMS 2023 Public Sector Operations Research (PSOR) Best Video Award.**
 - **Honorable Mention (third place) for CORS 2023 Practice Prize Competition.**
 - **Presented at MSOM 2023 Healthcare SIG.**
 - Preliminary version at The Journal of Nutrition.
16. **Pricing and Competition for Generative AI**
R. Mahmood*
Neural Information Processing Systems (NeurIPS), 37, 75727–75748, 2024.
 17. **Reasoning Paths with Reference Objects Elicit Quantitative Spatial Reasoning in Large Vision Language Models**
Y.-H. Liao*, R. Mahmood, S. Fidler, and D. Acuna
Empirical Methods in Natural Language Processing (EMNLP), 2024.
 18. **Prospective Human Validation of Artificial Intelligence Interventions in Cardiology: A Scoping Review**
A. Moosavi*, S. Huang*, M. Vahabi[†], B. Motamedivafa, N. Tian, R. Mahmood, P. Liu, and C. L. F. Sun
Journal of the American College of Cardiology (JACC): Advances, 2024.
 19. **Translating Labels to Solve Annotation Mismatches Across Object Detection Datasets**
Y.-H. Liao*[†], J. Lucas, R. Mahmood, V. Prabhu[†], D. Acuna, and S. Fidler
International Conference on Learning Representations (ICLR), 2024.
 20. **Bridging the Sim2Real Gap with CARE: Supervised Detection Adaptation with Conditional Alignment and Reweighting**
V. Prabhu*[†], D. Acuna, Y.-H. Liao[†], R. Mahmood, M. T. Law, J. Hoffman, S. Fidler, and J. Lucas
Transactions on Machine Learning Research (TMLR), 2023.
 21. **Optimizing Data Collection for Machine Learning**
R. Mahmood*, J. Lucas, J. M. Alvarez, S. Fidler, and M. T. Law
Neural Information Processing Systems (NeurIPS), 35, 29915–29928, 2022.
 22. **How Much More Data Do I Need? Estimating Requirements for Downstream Tasks**
R. Mahmood*, J. Lucas, D. Acuna, D. Li, J. Phillion, J. M. Alvarez, Z. Yu, S. Fidler, and M. T. Law
Computer Vision and Pattern Recognition (CVPR), 275–284, 2022.
 23. **Low Budget Active Learning via Wasserstein Distance: An Integer Programming Approach**
R. Mahmood*, S. Fidler, and M. T. Law
International Conference on Learning Representations (ICLR), 2022.
 24. **OpenKBP-Opt: An International and Reproducible Evaluation of 76 Knowledge-Based Planning Pipelines**
A. Babier*, R. Mahmood, B. Zhang, V. G. L. Alves, A. M. Barragán-Montero, J. Beaudry, C. E. Cardenas, Y. Chang, Z. Chen, J. Chun, K. Diaz, H. D. Eraso, E. Faustmann, S. Gaj, S. Gay, M. Gronberg, B. Guo, J. He, G. Heilemann, S. Hira, Y. Huang, F. Ji, D. Jiang, J. C. J. Giraldo, H. Lee, J. Lian, S. Liu, K. Liu, J. Marrugo, K. Miki, K. Nakamura, T. Netherton, D. Nguyen, H. Nourzadeh, A. F. I. Osman, Z. Peng, J. D. Q. Muñoz, C. Ramsel, D. J. Rhee, J. D. Rodriguez, H. Shan, J. V. Siebers, M. H. Soomro, K. Sun, A. U. Hoyos, C. Valderrama, R. Verbeek, E. Wang, S. Willems, Q. Wu, X. Xu, S. Yang, L. Yuan, S. Zhu, L. Zimmermann, K. L. Moore, T. G. Purdie, A. L. McNiven, and T. C. Y. Chan
Physics in Medicine & Biology, 67 (18), 2022.

25. **An Ensemble Learning Framework for Model Fitting and Evaluation in Inverse Linear Optimization**
 α - β A. Babier, T. C. Y. Chan, T. Lee, R. Mahmood*, and D. Terekhov
INFORMS Journal on Optimization (IJOO), 3 (2), 119–138, 2021.
 - Presented at CORS 2020 Canadian Healthcare Optimization Workshop.
 - **Honorable Mention (second place) for CORS 2018 Best Student Paper Competition.**
26. **Prediction of Protein and Fat Content in Human Donor Milk Using Machine Learning**
 R. K. Wong*[†], M. A. Pitino, R. Mahmood, I. Y. Zhu, D. Stone, S. Unger, D. L. O'Connor, and T. C. Y. Chan
The Journal of Nutrition, 2021.
27. **OpenKBP: The Open-access Knowledge-Based Planning Grand Challenge and Dataset**
 A. Babier*, B. Zhang, R. Mahmood, K. Moore, T. Purdie, A. McNiven, and T. C. Y. Chan
Medical Physics, 48 (9), 5549–5561, 2021.
28. **Sampling from the Complement of a Polyhedron: An MCMC Algorithm for Data Augmentation**
 α - β T. C. Y. Chan, A. Diamant, and R. Mahmood*
Operations Research Letters (ORL), 48 (6), 744–751, 2020.
29. **The Importance of Evaluating the Complete Knowledge-Based Planning Pipeline**
 A. Babier*, R. Mahmood, A. McNiven, A. Diamant, and T. C. Y. Chan
Physica Medica: European Journal of Medical Physics, 72, 73–79, 2020.
 - Preliminary version at ICCR 2019.
30. **AutoAudio: Deep Learning for Automatic Audiogram Interpretation**
 M. J. Crowson*, A. Hamour, R. Mahmood, A. Babier, V. Lin, D. Tucci, and T. C. Y. Chan
Journal of Medical Systems, 44 (163), 2020.
31. **Predicting Post-Operative Cochlear Implant Performance Using Supervised Machine Learning**
 M. J. Crowson*, P. Dixon, R. Mahmood, J. W. Lee, D. Shipp, T. Le, V. Lin, J. Chen, and T. C. Y. Chan
Otology & Neurotology, 41 (8), 1013–1023, 2020.
32. **The Importance of Evaluating the Complete Knowledge-based Automated Planning Pipeline**
 A. Babier*, R. Mahmood, A. McNiven, A. Diamant, and T. C. Y. Chan
International Conference on the Use of Computers in Radiotherapy (ICCR), 2019.
33. **Knowledge-based Automated Treatment Planning with Three-dimensional Generative Adversarial Networks**
 A. Babier*, R. Mahmood, A. McNiven, A. Diamant, and T. C. Y. Chan
Medical Physics, 47 (2), 297–306, 2019.
 - Presented at NeurIPS 2018 ML4H Workshop.
34. **Streaming Codes for Multiplicative-Matrix Channels with Burst Rank Loss**
R. Mahmood*, A. Badr, and A. Khisti
IEEE Transactions on Information Theory (IT), 64 (7), 5296–5311, 2018.
 - Preliminary version at ISIT 2016.
35. **Automated Treatment Planning in Radiation Therapy with Generative Adversarial Networks**
R. Mahmood*, A. Babier, A. McNiven, A. Diamant, and T. C. Y. Chan
Machine Learning for Healthcare (MLHC), PMLR 85, 484–499, 2018.

- **Runners' Up (second place) for CORS 2019 Health Care Operations Research (HCOR) Student Presentation Competition.**

36. **Convolutional Codes with Maximum Column Sum Rank for Network Streaming**

R. Mahmood*, A. Badr, and A. Khisti

IEEE Transactions on Information Theory (IT), 62 (6), 3039–3052, 2016.

- Preliminary version at ISIT 2015.

37. **Low Delay Network Streaming Under Burst Losses**

R. Mahmood*, A. Badr, and A. Khisti

IEEE International Symposium on Information Theory (ISIT), 2898–2902, 2016.

38. **Convolutional Codes with Maximum Column Sum Rank for Network Streaming**

R. Mahmood*, A. Badr, and A. Khisti

IEEE International Symposium on Information Theory (ISIT), 2271–2275, 2015.

39. **Embedded MDS Codes for Multicast Streaming**

A. Badr*, R. Mahmood, and A. Khisti

IEEE International Symposium on Information Theory (ISIT), 2276–2280, 2015.

Media Articles

40. **Optimizer for the 2021 NHL Expansion Draft**

M. Shin*, Y. Shalaby*, A. Loa*, B. Potter*, T. C. Y. Chan, and R. Mahmood

OR/MS Today, 48 (5), 52–54, 2021.

Patents

41. **Automated Data Curation for Autonomous Driving Planners via Error Reasoning**

N. Chang, M. Shen, R. Mahmood, and J. M. Alvarez Lopez

US Patent Application Number 19/702399, filed June 2026.

42. **Camera-Based Automatic Label Generation and Pruning**

J. Choi, R. Mahmood, M. T. Law, N. CHang, J. Lucas, and J. M. Alvarez Lopez

US Patent Application Number 19/694130, filed June 2026.

43. **Uncertainty Estimation for Object Detection in Autonomous and Semi-Autonomous Systems and Applications**

N. Durasov, J. Choi, R. Mahmood, M. T. Law, J. Lucas, and J. M. Alvarez Lopez

US Patent Number 2026/0111771 A1, published Apr 2026.

44. **Comparative Performance Assessment of Generative Artificial Intelligence Models**

R. Mahmood

US Patent Number 2025/0362953 A1, published Nov 2025.

45. **Using Neural Networks to Generate Bounding Boxes**

D. A. Marrero, R. Mahmood, J. Lucas, A. Liao, and S. Fidler

US Patent Number 2025/0086922 A1, published Mar 2025.

46. **Updating Synthetic Image Labels using Neural Networks to Improve Performance on Real-World Applications**

A. Liao, D. A. Marrero, J. Lucas, R. Mahmood, S. Fidler, and V. Prabhu

US Patent Number 2025/0054288 A1, published Feb 2025.

47. **Estimating Optimal Training Data Set Sizes For Machine Learning Model Systems And Applications**
R. Mahmood, J. Lucas, Z. Yu, J. M. Alvarez Lopez, S. Fidler, and M. T. Law
US Patent Number 2023/0376849 A1, published Nov 2023.
48. **Estimating Optimal Training Data Set Size For Machine Learning Model Systems And Applications**
R. Mahmood, J. Lucas, D. A. Marrero, D. Li, J. Philion, J. M. Alvarez Lopez, S. Fidler, and M. T. Law
US Patent Number 2023/0385687 A1, published Nov 2023.
49. **Optimized Active Learning Using Integer Programming**
R. Mahmood, S. Fidler, and M. T. Law
US Patent Number 2023/0244985 A1, published Aug 2023.

Presentations

Invited Seminars

University of Michigan MIDAS Seminar	2026
York University Center for AI and Society	2025
UTM Management Analytics Research Cluster (MARC) Conference	2025
ICCV Tutorial on Learning with Noisy and Unlabeled Data for Large Models beyond Categorization	2023
University of Toronto Rotman School of Management	2023
University of Ottawa Center for a Responsible Wealth Transition (CRWT)	2022
Wilfrid Laurier University Lazaridis School of Business and Economics	2022
University of Ottawa Telfer School of Management	2022
University of Hong Kong IMSE Department	2022
Rutgers University ISE Department	2021
University of North Carolina Kenan-Flagler Business School	2021
University of Cincinnati Lindner College of Business	2021
University of Iowa IE + EE Department	2021
University of Calgary CS Department	2021
University of Edinburgh Business School	2021
University of Alberta Alberta School of Business	2020
NVIDIA Toronto AI Lab	2020
University of Pittsburgh IE Department	2020
Université de Montréal GERAD	2019

Conference Talks²

Minority Reports: Balancing Cost and Quality in Ground Truth Data Annotation	
CORS Annual Conference, Kingston, ON, Canada	2026
INFORMS RMP Conference, New York City, NY, USA	2025
Pricing and Competition with Generative AI	
AIMOR Workshop, Banff, AB, Canada	2026
MSOM Conference, London, UK	2025
Training and Deploying Machine Learning Services under the Neural Scaling Law	
INFORMS Annual Meeting, Seattle, WA, USA	2024
Optimizing Data Collection for Machine Learning	
INFORMS Annual Meeting, Phoenix, AZ, USA	2023
INFORMS Workshop on Data Science, Phoenix, AZ, USA	2023
MSOM Conference, Montréal, QC, Canada	2023
Got (Optimal) Milk? Pooling Donations in Human Milk Banks with Machine Learning and Optimization	
MSOM Healthcare SIG Conference, Montréal, QC, Canada	2023
POMS Conference, Orlando, FL, USA	2023
Low Budget Active Learning: An Integer Programming Approach	
CORS Annual Conference, Vancouver, BC, Canada	2022
INFORMS Annual Meeting, Anaheim, CA, USA	2021
Learning to Optimize with Hidden Constraints	
POMS Conference, Orlando, FL, USA	2022
CORS Annual Conference, Toronto, ON, Canada	2021
INFORMS Annual Meeting, Washington, DC, USA	2020
INFORMS Annual Meeting, Seattle, WA, USA	2019
CORS Annual Conference, Saskatoon, SK, Canada	2019
An Ensemble Learning Framework for Inverse Linear Optimization	
INFORMS Health Care, Boston, MA, USA	2019
CORS Annual Conference, Saskatoon, SK, Canada	2019
CORS Annual Conference, Halifax, NS, Canada	2018
INFORMS Annual Meeting, Houston, TX, USA	2017
CORS Annual Conference, Quebec City, QC, Canada	2017
INFORMS Annual Meeting, Nashville, TN, USA	2016
Automated Treatment Planning with Generative Adversarial Networks	
CORS Annual Conference, Saskatoon, SK, Canada	2019
MLHC Conference, Palo Alto, CA, USA	2018
Convolutional Codes with Maximum Column Sum Rank for Network Streaming	
IEEE ISIT, Hong Kong, HK, China	2015

Teaching**University of Ottawa**

ADM 2304: Applications of Statistical Methods in Business	2023–2025
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²Presentations are categorized by the abbreviated main paper discussed. Actual titles may vary.

ADM 4307: Business Forecasting Analytics

2024–2025

MGT 5301: Predictive Analytics

2023–2025

Students Supervised**University of Ottawa PhD Students**

1. Maryam Vahabi, *Management* 2023–pres.
Co-advised with Christopher Sun.

University of Ottawa MSc Students

2. Viraj Vardhan, *Systems Engineering (Thesis)* 2025–pres.
Co-advised with Jonathan Li.
3. Yibo Wang, *Digital Transformation & Innovation (Thesis)* 2025–pres.
Co-advised with Justin Boutillier.
4. Morteza Emadi, *Business Analytics (Project)* 2024–2025
Co-advised with Michael Mulvey.
5. Hammad Shakir, *Business Analytics (Project)* 2024
6. Hsuan-Wei Liao, *Business Analytics (Project)* 2024
7. Tulika Tahiliani, *Business Analytics (Project)* 2024–2025

University of Ottawa Undergraduate Students

8. Joel Vadakken, *NSERC USRA* 2025
Visiting from the University of Toronto

University of Ottawa PhD Committee

9. Ziwei Zhang, *Digital Transformation & Innovation* 2025–pres.
10. Saeed Asadi, *Digital Transformation & Innovation* 2025–pres.
11. Shahryar Moradi, *Management* 2023–pres.
12. Amirhossein Moosavi, *Management* 2023

NVIDIA PhD Internships

13. Tolga Dimilioglu, *Research Scientist* 2025
Co-supervised with Nadine Chang, Maying Shen, and Jose M. Alvarez.
14. Feiyang Kang, *Research Scientist* 2024
Co-supervised with Nadine Chang, Maying Shen, Marc T. Law, James Lucas, and Jose M. Alvarez.
15. Nikita Durasov, *Research Scientist* 2024
Co-supervised with Jiwoong Choi, Marc T. Law, James Lucas, and Jose M. Alvarez.
16. Andrew Yuan-Hong Liao, *Research Scientist* 2022–2023
Co-supervised with David Acuna, James Lucas, and Sanja Fidler.
17. Viraj Prabhu, *Research Scientist* 2022
Co-supervised with David Acuna, Marc T. Law, James Lucas, and Sanja Fidler.

Grants

1. CIHR Project Grant, Co-Investigator, 2025–2030 (\$1 365 526).
2. CRWT Grant, Principal Investigator, 2025 (\$15 000).
3. CIHR Project Grant, Co-Investigator, 2024–2029 (\$1 450 725).
4. SSHRC Insight Development Grant, Co-Investigator, 2024–2026 (\$68 000).
5. NSERC Discovery Grant, Principal Investigator, 2023–2027 (\$160 000).
6. NSERC Discovery Grant ECR Launch Supplement, Principal Investigator, 2023–2024 (\$12 500).
7. University of Ottawa SEED Funding Opportunity, Principal Investigator, 2023 (\$20 000).
8. Telfer School of Management Start-up Grant, Principal Investigator, 2023-2024 (\$40 000).

Awards

1. Runners' Up, Innovative Applications in Analytics Award, INFORMS, 2024.
2. First Place, Pierskalla Best Paper Award, INFORMS, 2023.
3. Finalist, Public Sector Operations Research (PSOR) Best Video Award, INFORMS, 2023.
4. *(Declined)* Semi-Finalist, Wagner Prize Competition, INFORMS, 2023.
5. Finalist, Practice-Based Research Competition, MSOM, 2023.
6. Finalist, Practice Prize Award, CORS, 2023.
7. Runners' Up, College of Healthcare Operations Management (CHOM) Best Paper Prize, POMS, 2023 (\$250).
8. University of Toronto Doctoral Completion Award, 2019–2020 (\$8 000).
9. Runners' Up, Health Care Operations Research Student Presentation Competition, CORS, 2019.
10. Postgraduate Affiliate Award, Vector Institute for Artificial Intelligence, 2019 (\$12 000).
11. Honourable Mention, Student Paper Competition: Open Category, CORS, 2018 (\$100).
12. Postgraduate Doctoral Scholarship, NSERC, 2017 (\$42 000).
13. First Place, Waterfront International Ltd. Quantathon, 2016 (\$7 500).

Service

Organizational Service for the Academic Community

NeurIPS	2026
Area Chair	
ICLR	2026
Area Chair	

CVPR Workshop on Exploring the Next Generation of Data (NexD)	2025
Workshop Organization Committee Member	
INFORMS HAS Pierskalla Award Committee	2024
Co-Chair	

Refereeing

Grants

SSHRC Insight Grant (External Reviewer) 2024–2025

Journals

Manufacturing & Services Operations Research; Operations Research; INFORMS Journal on Computing; European Journal of Operational Research; Computers and Operations Research; Health Care Management Science; IEEE Transactions on Knowledge and Data Engineering; IEEE Transactions on Cybernetics; IIE Transactions

Conferences

MSOM 2026; ICLR 2022–2025; NeurIPS 2020–2025; ICML 2021–2025; IEEE ISIT 2017, 2022; ACM CHIL 2020–2021; NeurIPS ML4H Workshop 2019–2020

Service for the University of Ottawa

Telfer Business Healthcare Society	2023–pres.
Faculty Advisor	
University of Ottawa UCaaS	2022–2023
Steering Committee Member	

Other Professional Experience

NHL Expansion Draft Optimizer	2017, 2021
http://nhlexpansiondraft.com	
Back-end Software Developer (2017), Adviser (2021)	

We deployed a web app simulating the 2017 and 2021 NHL Expansion Drafts. Our site was featured in *The Toronto Star*, *OR/MS Today*, and *The Seattle Times*.

OpenKBP Grand Challenge	2019–2020
https://www.aapm.org/GrandChallenge/OpenKBP/	
Machine Learning Expert	

We organized an international competition for automating radiation therapy dose treatments in head-and-neck cancer, featuring 28 teams of 195 participants. We also released the public-access OpenKBP Data Set containing 400 treatments.

Opus One Solutions, Toronto, ON, Canada	2019
Power Systems Optimization Expert (Consultant)	

Personal

Languages: English (fluent), French (beginner)

Citizenship: Canadian

Last updated: July 6, 2026
<http://rafidrm.github.io>